



Intelligence Academy

Scikit-Learn 101

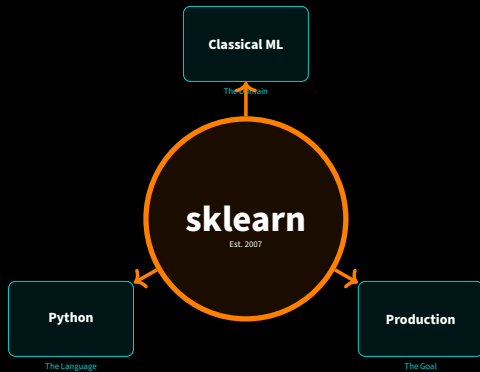
01. Introduction: What is Scikit-Learn

Mejbah Ahammad

The Foundation & Ecosystem

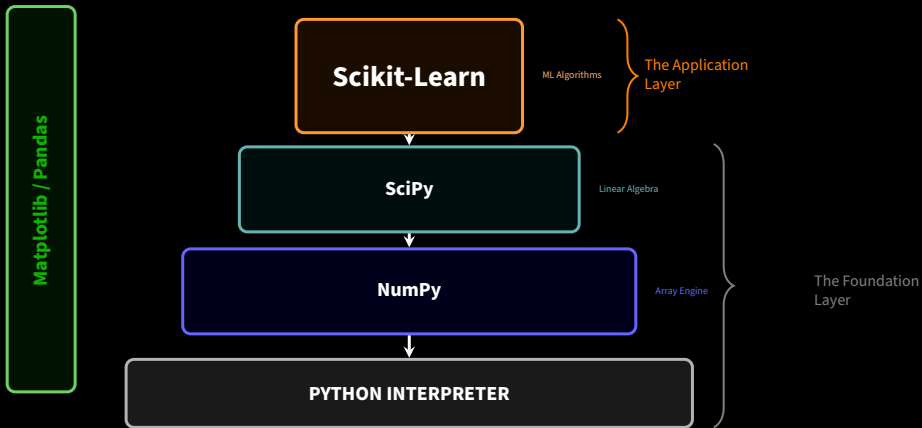
1.1 The Industry Standard

Scikit-learn is not just a library; it is the **Gold Standard** for Classical Machine Learning in Python. While Deep Learning has PyTorch, for everything else (Regression, Classification, Clustering), the world runs on Scikit-learn.



1.2 The SciPy Stack Hierarchy

Scikit-learn does not stand alone. It stands on the shoulders of giants. It relies on **NumPy** for high-performance array math and **SciPy** for fundamental scientific algorithms.



1.3 Key Characteristics

Why is it safe for your company to use?

- **Open Source:** Transparent and free.
- **BSD License:** Commercially usable (unlike GPL).
- **Community:** Backed by thousands of contributors.

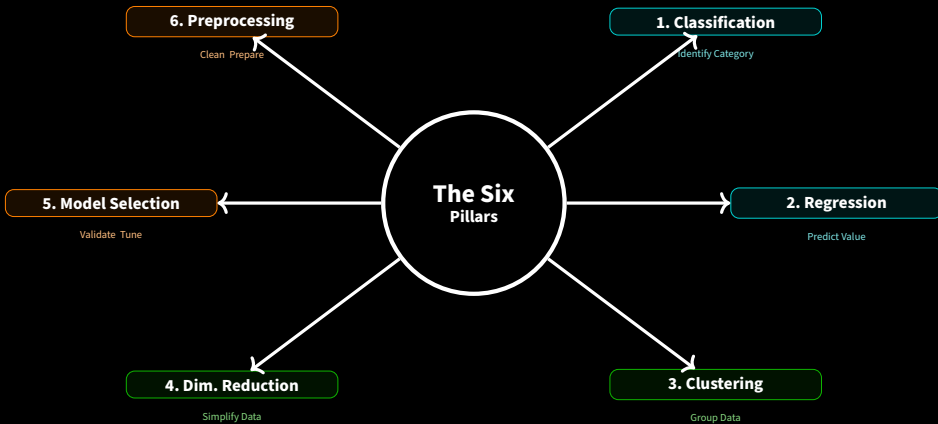


The Pillars of Trust

The Six Pillars of Functionality

2.1 The Functional Overview

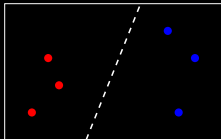
Scikit-learn isn't just one tool; it's a Swiss Army Knife with six distinct blades. We can categorize them into **Supervised Learning**, **Unsupervised Learning**, and **Infrastructure**.



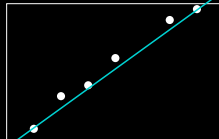
2.2 Supervised vs. 2.3 Unsupervised

Classification draws lines to separate things. **Regression** draws lines to follow trends. **Clustering** finds blobs. **Dim Reduction** squashes data.

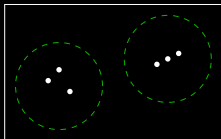
Classification



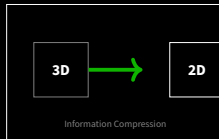
Regression



Clustering

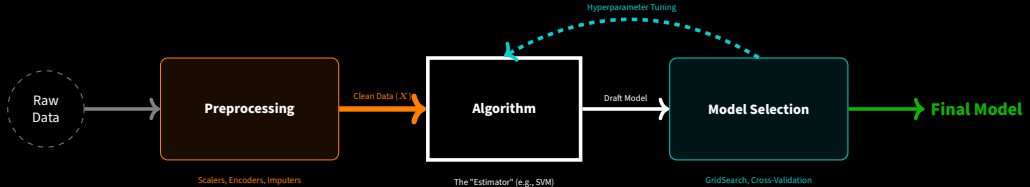


Dim. Reduction



2.5 & 2.6 The Support Infrastructure

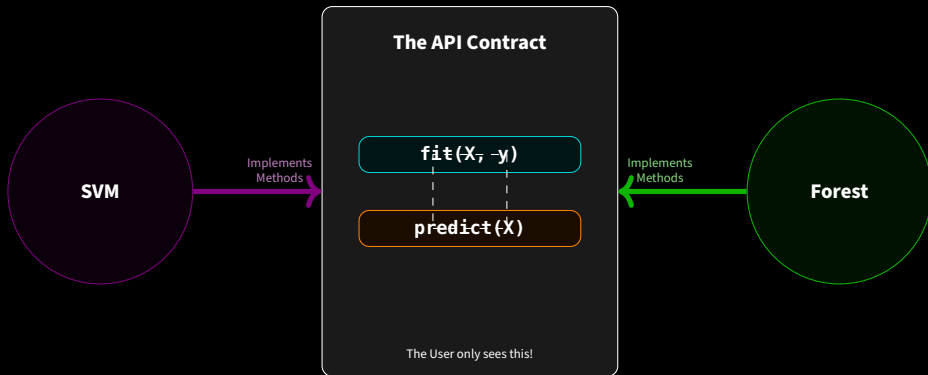
These pillars wrap around your model. **Preprocessing** happens *before* the model sees data. **Model Selection** happens *after/during* training to ensure quality.



The Design Philosophy

3.1 Consistency: The Universal Interface

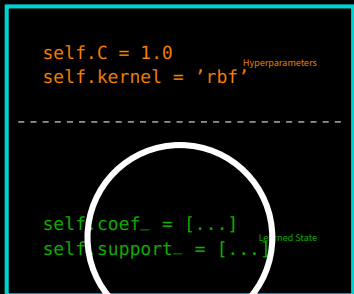
In scikit-learn, the syntax is predictive. If you know how to use one model, you know how to use them all. This "Duck Typing" approach reduces cognitive load.



3.2 Inspection: The Glass Box Model

Scikit-learn models are not "Black Boxes." They are transparent. Anything the model learns is stored in public attributes ending with an underscore (e.g., `coef_`).

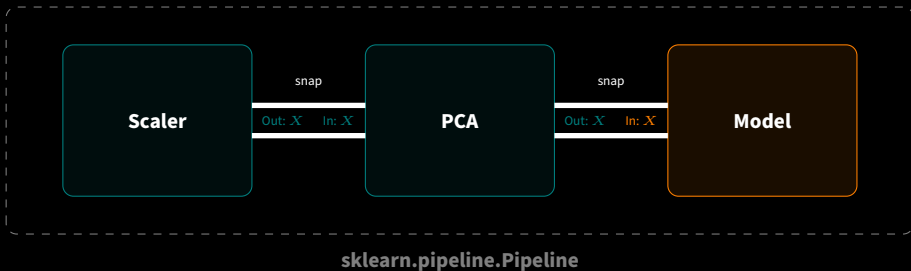
The Estimator



Accessible via `vars(model)` or `model.attribute_`

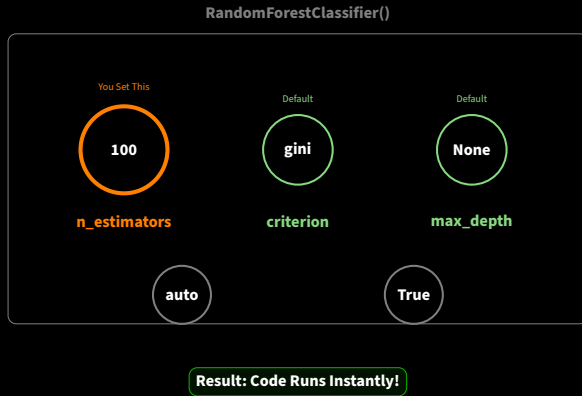
3.3 Composition: Building Pipelines

Because interfaces are consistent, we can snap blocks together to build complex workflows without writing "glue code."



3.4 Sensible Defaults

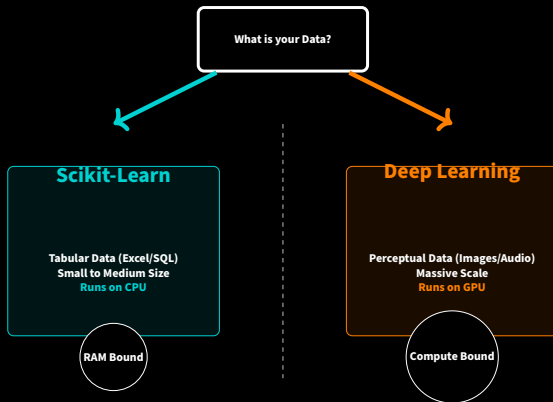
Models have dozens of mathematical knobs. Scikit-learn sets them to "Safe Defaults" so you can run a model immediately. You only tune what you need.



Scikit-learn vs. The Rest

4.1 The Hardware Divide: Classical vs. Deep

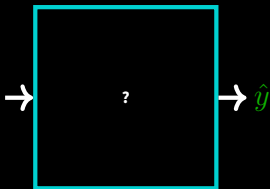
When should you leave Scikit-learn and move to PyTorch or TensorFlow? It comes down to **Data Structure** and **Hardware**.



4.2 The Goal: Engineer vs. Scientist

Scikit-learn is for Engineers: "Does it work?" (Prediction Accuracy). **Statsmodels** is for Scientists: "Why does it work?" (Statistical Inference/P-values).

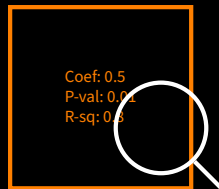
Scikit-Learn



Focus: Accuracy
(The Result)

VS

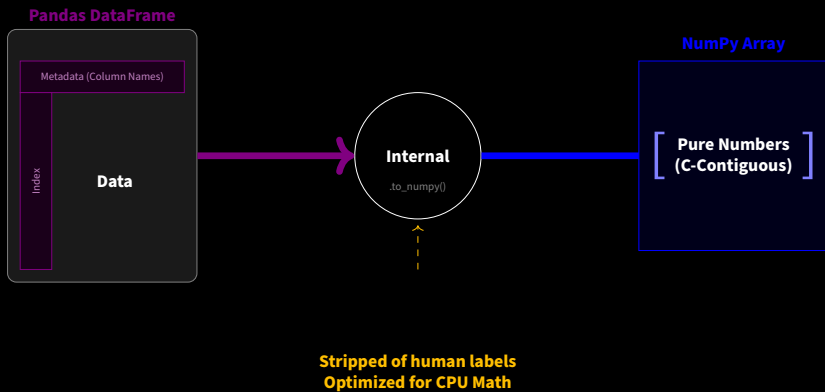
Statsmodels



Focus: Interpretability
(The Process)

4.3 Handling Data: The Ingestion Layer

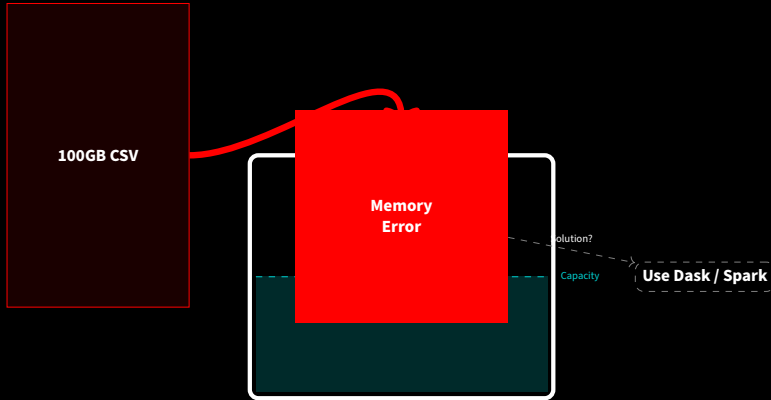
Scikit-learn loves Math (Matrices), not Tables. While it *accepts* Pandas DataFrames, it immediately strips away the column names and indices, converting everything into raw NumPy arrays for calculation.



Limitations & Boundaries

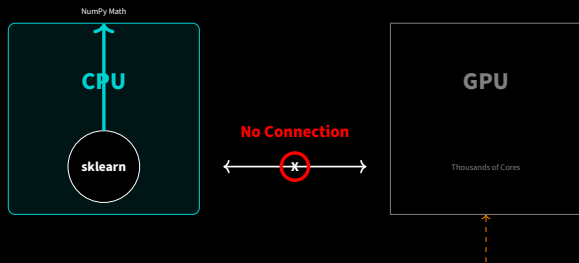
5.1 The RAM Ceiling

Scikit-learn requires the *entire* dataset to sit in RAM at once. It does not support "Out-of-Core" learning (streaming data from disk) for most algorithms. **The Rule:** If Data > RAM, Scikit-learn crashes.



5.2 The CPU-Only Architecture

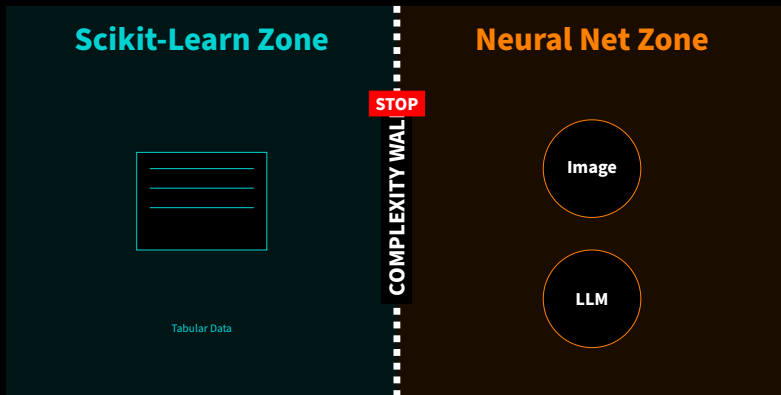
Scikit-learn made a conscious design choice to avoid GPU support. Why? To keep the library easy to install (no CUDA drivers hell) and easy to maintain. It relies on NumPy, which is CPU-bound.



Need Speed?
Use RAPIDS (cuML)

5.3 Not for Deep Learning

Scikit-learn works on **Structured Data** (Excel). It does not understand **Perceptual Data** (Images, Text, Audio). Do not try to build an LLM or a Face Detector with it.



End of Lecture: What is Scikit-Learn?

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